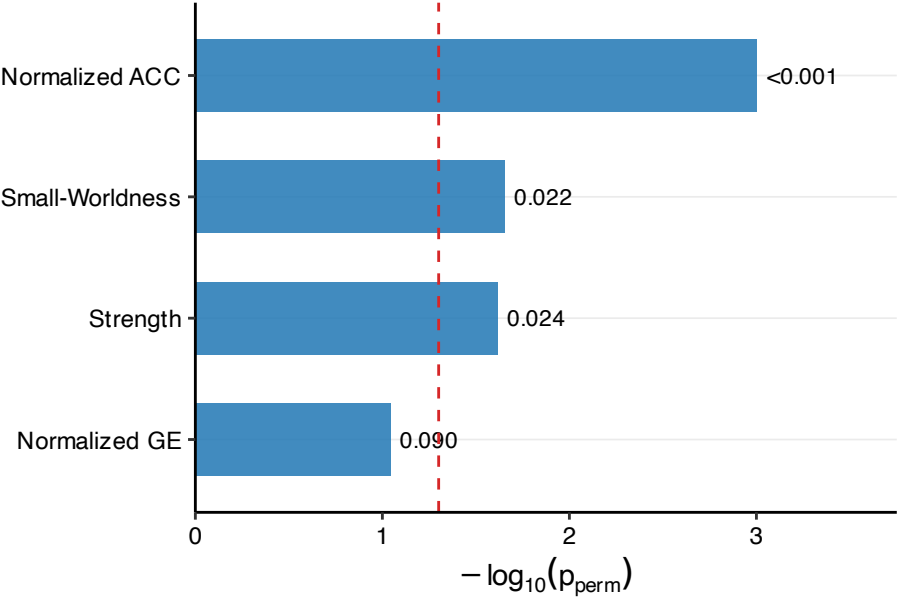


Supplementary Figure 20. Permutation-based exposure-metric association tests

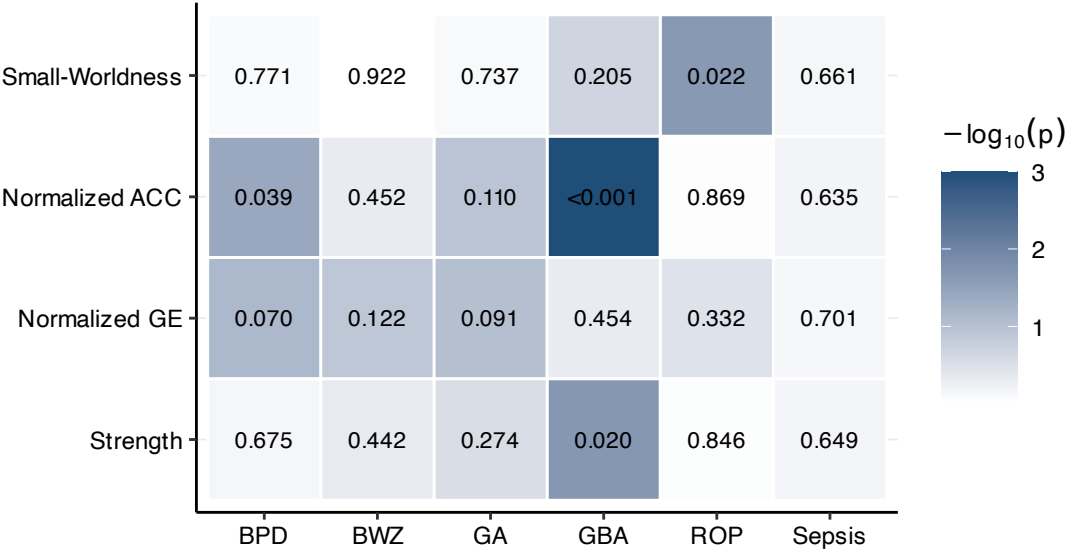
A. Global exposure set

Joint association of all exposures with FPCA scores



B. Incremental (per-exposure)

Each exposure's unique contribution beyond the others



C. Latent severity (PC1)

Single-axis exposure summary

